Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627; www.cwsei.ubc.ca/resources/COPUS.h

Date: _____ **Class:** _____ **Instructor:** _____ **No. students** _____ **Observer Name:** _____

Classroom arranged how? (include any special notes about environment) _____

1. **L**-Listening; **Ind**-Individual work; Fol - Follow-along worksheet; **CG**-Clicker Q discussn; **WG**-Worksheet group work; **C**-Writing code on computer; **OG**-Other group work; **AnQ**-Answer Q; **SQ**-Student Q; **WC**-Whole class discuss.; **Prd**-Predicting; **SP**-Student present.; **TQ**-
2. **Lec**-Lecturing; **RtW**-Writing; **FUp**-Follow-up; **PQ**-Pose Q; **CQ**-Clicker Q; **AnQ**-Answer Q; **MG**-Moving/Guiding; **1o1**-One-on-one; **D/V**-Demo+; **Adm**-Admin; **W**-Waiting; **O**-Other
3. **L** - low engagement (observer confident majority of students are not engaged); **M** - medium engagement (observer is unsure); **H** - high engagement (observer is confident majority of students are engaged); **InterI** - number of unique students instructor interacted with; **InterT** -

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

	1. Students doing															2. instructor doing												3. Engagement					Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
0 - 2																																	
2																																	
4																																	
6																																	
8 - 10																																	
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
10 - 12																																	
12																																	
14																																	
16																																	
18 - 20																																	
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
20 - 22																																	
22																																	
24																																	
26																																	
28 - 30																																	

	1. Students doing															2. instructor doing												3. Engagement					Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
30 - 32																																
32																																
34																																
36																																
38 - 40																																
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
40 - 42																																
42																																
44																																
46																																
48 - 50																																
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
50-52																																
52																																
54																																
56																																
58-60																																

	1. Students doing															2. instructor doing												3. Engagement					Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
60-62																																
62																																
64																																
66																																
68-70																																
	L	Ind	Fol	CG	WG	C	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	InterI	InterT	
70-72																																
72																																
74																																
76																																
78-80																																
Summary																																